Page 2: Introduction – What are Acids and Bases?

In our daily life, we come across many substances that have distinct tastes and textures. These are often classified as acids or bases based on their chemical nature.

What is an Acid?
The word 'acid' comes from the Latin word 'acidus' meaning 'sour'.

- **Definition (Arrhenius Concept):** An acid is a substance that dissociates in water to produce hydrogen ions (H^+).
- **Examples:**
  - **Natural Acids:** Found in food items.
    - Citric acid (Lemons, Oranges)
    - Acetic acid (Vinegar)
    - Lactic acid (Curd, Yogurt)
    - Tartaric acid (Tamarind, Grapes)
    - Oxalic acid (Tomatoes)
  - **Mineral Acids:** Prepared from minerals.
    - Hydrochloric acid (HCl)
    - Sulphuric acid (H_2SO_4)
    - Nitric acid (HNO_3)

What is a Base?

- **Definition (Arrhenius Concept):** A base is a substance that dissociates in water to produce hydroxide ions (OH^-).
- **Examples:**
  - Sodium hydroxide (NaOH) – Caustic soda
  - Calcium hydroxide ($Ca(OH)_2$) – Lime water
  - Potassium hydroxide (KOH) – Caustic potash
  - Magnesium hydroxide ($Mg(OH)_2$) – Milk of magnesia
  - Ammonium hydroxide (NH_4OH)

Alkalis: Bases that are soluble in water are called **alkalis**. All alkalis are bases, but not all bases are alkalis (e.g., copper oxide is a base, but it is insoluble in water, so it is not an alkali).

Page 3: Properties of Acids and Bases

How do we know if a substance is an acid or a base? They have characteristic properties.

Physical and Chemical Properties of Acids:

1. **Taste:** They have a sour taste. (NEVER taste chemicals in the lab!)
2. **Touch:** They are usually corrosive and can cause burns on the skin.
3. **Effect on Indicators:** They turn blue litmus paper **red**.
4. **Reaction with Metals:** Acids react with active metals (like Zn, Mg, Al) to produce **hydrogen gas**. The presence of hydrogen gas can be tested by bringing a burning candle near it – it burns with a 'pop' sound.
   - *Example:* $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$
5. **Reaction with Metal Carbonates and Bicarbonates:** Acids react with carbonates and bicarbonates to produce **carbon dioxide gas**, which turns lime water milky.
   - *Example:* $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2\uparrow$
   - *Example:* $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2\uparrow$
6. **Reaction with Bases:** They react with bases to form salt and water. This is called **neutralization**.

Physical and Chemical Properties of Bases:

1. **Taste:** They have a bitter taste.
2. **Touch:** They feel soapy or slippery to the touch (like soap solution).
3. **Effect on Indicators:** They turn red litmus paper **blue**.
4. **Reaction with Metals:** Some bases (like NaOH, KOH) react with metals like Al and Zn to produce hydrogen gas.
5. **Reaction with Acids:** They react with acids to form salt and water (neutralization).

Page 4: Indicators – The Taste-Safe Testers

Since tasting and touching chemicals is dangerous, we use special substances called **indicators** to test whether something is acidic or basic. Indicators change colour depending on the nature of the substance.

Types of Indicators:

Type of Indicator	Example	Colour in Acid	Colour in Base
Natural Indicators	Litmus (from Lichen)	**Red**	**Blue**
	Turmeric	Yellow	**Reddish Brown**
	Red Cabbage	**Reddish**	**Greenish Yellow**
Synthetic Indicators	Phenolphthalein	**Colourless**	**Pink**
	Methyl Orange	**Red / Pink**	**Yellow**
Olfactory Indicators	Onion	Smell remains (acid)	Smell destroyed (base)
	Vanilla	Smell remains (acid)	Smell destroyed (base)

Olfactory Indicators: These are substances whose smell changes in acidic or basic solutions. They are very useful for visually impaired students during experiments.

Page 5: The pH Scale – Measuring Strength

Not all acids and bases are equally strong. Some are very strong and corrosive, while others are weak and safe to consume. The **pH scale** is used to measure how acidic or basic a substance is.

What is pH?
pH stands for "Potential of Hydrogen". It is a number that indicates the concentration of H^+ ions in a solution.

- The scale ranges from **0 to 14**.

Understanding the Scale:

- **pH = 7:** The solution is **neutral** (e.g., pure water).
- **pH < 7:** The solution is **acidic**. Lower the pH, stronger the acid.

- **pH > 7:** The solution is **basic (alkaline)** . Higher the pH, stronger the base.

Importance of pH in Daily Life:

1. **Our Digestive System:** Our stomach produces hydrochloric acid (pH ~1-2) to digest food. Excess acid causes "acidity" or heartburn. **Antacids** (like milk of magnesia, which is basic) are taken to neutralize this excess acid.
2. **Soil pH:** Plants grow best at a specific pH. If soil is too acidic (low pH), farmers add **slaked lime (calcium hydroxide)** or chalk to neutralize it. If soil is too basic, they add organic matter (compost).
3. **Tooth Decay:** Bacteria in our mouth break down food particles to produce acids. This lowers the pH in the mouth, attacking tooth enamel (which is made of calcium phosphate). Using **toothpaste** (which is basic) helps neutralize this acid.
4. **Bee Stings / Nettle Stings:** Bee stings are acidic (formic acid). They can be treated with a mild base like **baking soda** (sodium hydrogen carbonate). Nettle stings are also acidic.
5. **Wasp Stings:** Wasp stings are basic. They are treated with a mild acid like **vinegar**.

Page 6: Reaction of Acids and Bases with Each Other – Neutralization

Neutralization Reaction:
When an acid and a base react with each other, they cancel out each other's effects. The products are **salt and water**. This reaction is exothermic (releases heat).

Acid + Base → Salt + Water
Example: $HCl + NaOH \rightarrow NaCl + H_2O$

Reaction of Acids with Metal Oxides:
Metal oxides are generally basic in nature. When an acid reacts with a metal oxide, it also forms salt and water. This is similar to a neutralization reaction.

Metal Oxide + Acid → Salt + Water
Example: CuO (Copper oxide - black) + $H_2SO_4 \rightarrow CuSO_4$ (Copper sulphate - blue) + H_2O

Reaction of Bases with Non-metal Oxides:
Non-metal oxides are generally acidic in nature. When a base reacts with a non-metal oxide, it forms salt and water.

Non-metal Oxide + Base → Salt + Water
Example: CO_2 (Carbon dioxide) + $Ca(OH)_2$ (Lime water) → $CaCO_3$ (Calcium carbonate - white precipitate) + H_2O

(This reaction is used to test for CO_2 gas – lime water turns milky due to the formation of insoluble $CaCO_3$.)

Page 7: What are Salts? – The Family of Salts

A salt is an ionic compound formed by the neutralization of an acid and a base.

General Concept:

- The **cation (positive ion)** comes from the **base**.
- The **anion (negative ion)** comes from the **acid**.

Example: In NaCl (Sodium Chloride)

- Na^+ comes from the base NaOH.
- Cl^- comes from the acid HCl.

Family of Salts:
Salts that have the same positive ions or negative ions belong to a family.

1. **Chlorides:** Salts with Cl^- ion (e.g., NaCl, KCl, NH_4Cl).
2. **Sulphates:** Salts with SO_4^{2-} ion (e.g., $CuSO_4$, Na_2SO_4).
3. **Nitrates:** Salts with NO_3^- ion (e.g., KNO_3, $AgNO_3$).
4. **Carbonates:** Salts with CO_3^{2-} ion (e.g., Na_2CO_3, $CaCO_3$).

Types of Salts based on pH:

1. **Normal Salts:** Formed by complete neutralization. They are neutral (pH = 7). *Example: NaCl, KNO_3.*
2. **Acidic Salts:** Formed by partial neutralization of a polybasic acid (an acid with more than one H^+ ion). They have a replaceable H^+ ion and are acidic (pH < 7). *Example: $NaHSO_4$ (Sodium hydrogen sulphate).*
3. **Basic Salts:** Formed by partial neutralization of a polyacidic base (a base with more than one OH^- ion). They have a replaceable OH^- ion and are basic (pH > 7). *Example: Mg(OH)Cl (Basic magnesium chloride).*

Page 8: Common Salt – A Vital Chemical

Sodium Chloride (NaCl)

- **Source:** It is obtained from seawater (by evaporation) or from rock salt deposits.

- **Common Name:** Common salt, table salt.
- **Use:** It is essential for our diet, used for preserving food (pickling), and is a raw material for many chemicals.

Important Chemicals from Common Salt:

1. **Caustic Soda (Sodium Hydroxide - NaOH)**
    - **Production:** Manufactured by the **Chlor-Alkali process**. Electricity is passed through an aqueous solution of sodium chloride (brine).
    - **Reaction:** $2NaCl\ (aq) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + Cl_2\ (g) + H_2\ (g)$
    - **Products:** This process yields three important products: NaOH, Chlorine gas, and Hydrogen gas. (The name 'Chlor-Alkali' comes from **Chlor**ine and **Alkali**).
    - **Uses of NaOH:** It is a strong base used in making soaps, detergents, paper, textiles, and refining petroleum.
2. **Chlorine Gas (Cl_2)**
    - **Uses:**
        - Used for water purification (to kill germs).
        - Used in the manufacture of bleaching powder, PVC (polyvinyl chloride), pesticides, and CFCs.
        - Used in the production of hydrochloric acid.
3. **Hydrogen Gas (H_2)**
    - **Uses:**
        - Used as a fuel (hydrogen fuel cells).
        - Used in the manufacture of ammonia (Haber's process for fertilizers) and hydrochloric acid.
        - Used in the hydrogenation of oils to make vanaspati ghee.

Page 9: Some Important Compounds and Their Uses

1. Bleaching Powder (Calcium Oxychloride - $CaOCl_2$)

- **Production:** Produced by the action of chlorine gas on dry slaked lime [$Ca(OH)_2$].
    - $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$
- **Properties:** It smells like chlorine. It is a strong oxidizing agent.

- **Uses:**
    - For bleaching cotton and linen in the textile industry.
    - For bleaching wood pulp in paper mills.
    - As an oxidizing agent in many industries.
    - For disinfecting drinking water to make it germ-free.

2. Baking Soda (Sodium Hydrogen Carbonate - $NaHCO_3$)

- **Production:** Manufactured using sodium chloride, ammonia, and carbon dioxide (Solvay process).
    - $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$
- **Properties:** A mild, non-corrosive base. On heating, it decomposes to produce sodium carbonate, CO_2, and water.
- **Uses:**
    - In baking (as a leavening agent). When heated, it produces CO_2 gas, which makes cakes and bread fluffy.
    - In antacids to relieve acidity.
    - In soda-acid fire extinguishers (it reacts with acid to produce CO_2).

3. Washing Soda (Sodium Carbonate Decahydrate - $Na_2CO_3 \cdot 10H_2O$)

- **Production:** Obtained by heating baking soda (recrystallization).
    - $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$
    - The Na_2CO_3 is then recrystallized to get washing soda ($Na_2CO_3 \cdot 10H_2O$).
- **Uses:**
    - In glass, soap, and paper industries.
    - For removing permanent hardness of water.
    - As a cleaning agent.

4. Plaster of Paris (Calcium Sulphate Hemihydrate - $CaSO_4 \cdot ½H_2O$)

- **Production:** Prepared by heating gypsum (Calcium sulphate dihydrate) to 373 K (100°C).
    - $CaSO_4 \cdot 2H_2O$ (Gypsum) $\rightarrow CaSO_4 \cdot ½H_2O$ (Plaster of Paris) + 1½ H_2O

- **Properties:** When mixed with water, it sets into a hard, solid mass (gypsum). It expands slightly on setting, making it ideal for casts.
- **Uses:**
    - For making plaster casts for fractured bones.
    - In dentistry and sculpture.
    - For making decorative materials (false ceilings, idols).

Page 10: Water of Crystallization and Revision Summary

Water of Crystallization
Many salts contain water molecules loosely bonded to their crystal structure. This water is called **water of crystallization**. The salts are called **hydrated salts**.

Hydrated Salt	Chemical Formula	Common Name	Colour	Water Molecules
Copper Sulphate	$CuSO_4 \cdot 5H_2O$	Blue Vitriol	**Blue**	5
Ferrous Sulphate	$FeSO_4 \cdot 7H_2O$	Green Vitriol	**Light Green**	7
Sodium Carbonate	$Na_2CO_3 \cdot 10H_2O$	Washing Soda	White	10
Calcium Sulphate	$CaSO_4 \cdot 2H_2O$	Gypsum	White	2

Effects on Hydrated Salts:

- **Heating:** On heating, hydrated salts lose their water of crystallization and become **anhydrous** (without water). Their colour may change.
    - *Example:* Blue $CuSO_4 \cdot 5H_2O$ on heating becomes white anhydrous $CuSO_4$. Adding water restores the blue colour.

Important Phenomena:

- **Efflorescence:** When hydrated salts are exposed to air, they lose their water of crystallization and become powdery. (e.g., Washing soda).
- **Hygroscopy:** Substances that absorb moisture from the air but do not dissolve in it. (e.g., Plaster of Paris, anhydrous $CaCl_2$).
- **Deliquescence:** Substances that absorb enough moisture from the air to dissolve in it and form a solution. (e.g., Calcium chloride, NaOH pellets, P_2O_5).

Revision Table:

Concept	Key Point
Acid	Turns blue litmus red; gives H^+ ions in water; sour taste.
Base	Turns red litmus blue; gives OH^- ions in water; bitter, soapy touch.
Alkali	A water-soluble base.
pH Scale	0-14; <7 Acid, 7 Neutral, >7 Base.
Neutralization	Acid + Base → Salt + Water (Exothermic).
Common Salt (NaCl)	Raw material for NaOH, Cl_2, H_2 via Chlor-Alkali process.
Bleaching Powder ($CaOCl_2$)	For bleaching and disinfection.
Baking Soda ($NaHCO_3$)	For baking, antacids, fire extinguishers.
Washing Soda ($Na_2CO_3 \cdot 10H_2O$)	For cleaning and water softening.
Plaster of Paris ($CaSO_4 \cdot ½H_2O$)	For casts and sculptures (sets on adding water).
Water of Crystallization	Fixed number of water molecules in a crystal structure.